

Module 4: Cells - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Which statement is part of cell theory?
2. A. Cells arise only from nonliving matter.
3. B. All organisms are made of cells, and cells come from existing cells.
4. C. Cells are larger than the organisms they build.
5. D. Only animals are made of cells.
6. Answer: B
7. Why: Cell theory states that organisms are made of cells and that cells arise from pre-existing cells.
8. What is the clearest structural difference between a prokaryotic and a eukaryotic cell?
9. A. Only prokaryotes contain DNA.
10. B. Only eukaryotes are alive.
11. C. Prokaryotes are always larger.
12. D. Eukaryotes have a membrane-bound nucleus; prokaryotes do not.
13. Answer: D
14. Why: The defining feature is the membrane-bound nucleus (and other organelles) found in eukaryotes but not prokaryotes.
15. A cell is packed with mitochondria. This most likely means the cell:
16. A. has a high demand for usable energy
17. B. does not use any energy
18. C. performs only photosynthesis
19. D. contains no DNA
20. Answer: A
21. Why: Mitochondria harvest usable energy, so energy-hungry cells such as muscle cells contain many of them.

22. Which structure would you find in a plant cell but not an animal cell?

23. A. Nucleus

24. B. Mitochondrion

25. C. Chloroplast

26. D. Cell membrane

27. Answer: C

28. Why: Chloroplasts carry out photosynthesis and are found in plants and algae, not in animal cells.